SidPrep 2.0

Complete Rebuild Plan | Student App + Public Website + Admin Panel + Shared Backend

Rebuild SidPrep from a clean starting point with one lightweight student mobile application, a public student website and one professional administration panel. The old repository is a business-requirements reference, not the codebase to redesign.

1. Recommended architecture

STUDENT MOBILE APP **Android / iOS**	**PUBLIC WEBSITE + ADMIN** **sidprep.in and /admin**
SHARED API **api.sidprep.in /v1**	**CONTENT ENGINE** **Scheduled fetch / approval**
Oracle Cloud Fastify + PostgreSQL + job queue	Cloudflare (DNS, CDN, R2) + Brevo (email OTP)

Use **one monorepo, one database and one shared backend API**. Separate backends for the app, website and admin panel would add cost and operational complexity.

Recommended technology

Component	Technology
Student mobile application	React Native + Expo + TypeScript
Public student website	Next.js, static-first public pages
Admin panel	Next.js + Tailwind CSS + shadcn/ui
Shared API	Fastify + TypeScript
Database and scheduled jobs	PostgreSQL + self-hosted cron / PostgreSQL queue
Files	Cloudflare R2 within applicable free limits
Email authentication	Email OTP through Brevo
Server	Existing Oracle Cloud deployment
Protection and logs	Cloudflare Free + application security; PostgreSQL audit events + rotated server logs

2. The reality of free hosting

The earlier Oracle document assumed 4 OCPUs and 24 GB RAM. The current free allocation described in the prior answer was equivalent to 2 Ampere A1 OCPUs and 12 GB RAM. Check resources available in your actual tenancy; free-tier capacity, eligibility and limits can change. Oracle warns that idle instances may be reclaimed.

Service	Allowance / limitation noted in the plan
Oracle Cloud	Eligible compute and storage within its current Always Free allowance
Cloudflare Pages	500 builds per month
Cloudflare R2	10 GB-month of standard storage; operations are limited
Brevo	300 emails per day
Expo EAS	Limited free cloud builds
Google Play	US$25 one-time developer registration fee
Apple App Store	US$99/year membership

Practical constraint: A zero-monthly-hosting-cost beta is possible, but unlimited high traffic at guaranteed zero cost is not. Monitor quotas, place limits on storage and email, and scale when actual usage requires it. Domain registration/renewal is separate.

3. Two complete implementation prompts

Prompt 1 - SidPrep Student Mobile App: React Native + Expo; Android/iOS; exam preparation; study plans; mocks; current affairs; in-app PDF/DOCX/image viewing; offline handling; telemetry and diagnostics.

Prompt 2 - Website + Admin Panel + Backend: student-facing website and SEO; exam management; auto-news ingestion; editorial workflows; document library; accounts and admin roles; audit logs; backend APIs and Oracle deployment.

The complete master prompts are supplied as separate Markdown downloads alongside this PDF. Their recommended execution order is Website/Admin/Backend first, then Mobile App.

4. Automated current affairs

Current affairs should be exam-focused: relevance and tags should distinguish SSC, RRB, WBPSC and IBPS. Sources must be permitted and verifiable; publish short original summaries with attribution, rather than reproducing full copyrighted passages.

Example time (IST)	Scheduled workflow
5:30 AM	Collect permitted RSS/API stories, such as relevant official PIB sources, then normalize and deduplicate.
6:00 AM	Prepare original draft summaries and assign relevant exam, topic and region tags.
Before 7:00 AM	Admin reviews each article, checks sources and approves, edits or rejects.
7:00 AM	Publish approved stories to app and website; deliver opted-in notifications within available quotas.

Publication modes: (1) **Manual approval**, the safer default: no approval means no publication. (2) **Conditional automatic publishing**, off by default: only specifically allowlisted, policy-compliant trusted sources/categories with strict quality checks; sensitive or uncertain information still requires editorial review. Log each decision, and support corrections or withdrawals.

5. In-app and browser file viewing

Admin control	Student result
Preview allowed	View authorized PDF, sanitized DOCX preview or image inside app/website.
Download allowed	Show Download action and authorize the underlying download request.
Download disabled	Hide the button and deny direct download access.
Exam-specific visibility	Restrict materials by audience and exam.
Expiry or revoke	Stop future access when the configured policy changes.
Activity tracking	Record authorized view and download events.

For DOCX, generate safe previews on the server at upload time. A preview-only restriction is **not** screenshot-proof DRM.

6. Better administration and user management

Student management: registration, selected exams, attempts, progress, account status, active sessions and support requests.

Content activity: creator, edits, approvals, scheduled publication, corrections, version changes and withdrawal history.

System activity: API errors, scheduled job failures, Brevo quota/delivery, storage errors, app crashes and admin security events.

Apply role-based access, review/approval permissions, retention limits and audit records to sensitive actions. Avoid collecting secrets or excessive personal information in logs.

7. Recommended development sequence

Stage 1. Back up and archive the old project; create a clean replacement repository.

Stage 2. Implement database, shared API, website and admin panel foundation.

Stage 3. Build the student app against the same API contract.

Stage 4. Integrate current-affairs automation, document library and logging.

Stage 5. Run functional, security and cost/usage tests; deploy after explicit approval.

Final recommendation

Do not delete the old repository or production configuration immediately. Archive it, create SidPrep 2.0 separately, test the replacement, and retire the legacy deployment only after the new service is operational. Prioritize stable core workflows and free-tier limits over unnecessary microservices or decorative features.

Reference material: earlier uploaded SidPrep specifications and architecture documents; original master prompts are separate supporting files. Live service limits should be reconfirmed before production launch.